

First Assignment: Image Stitching

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1 Introduction

This report presents my work for the panorama stitching pipeline assignment, combining Harris corner detection, normalized patch-based descriptors, and feature matching using both Euclidean distance and normalized cross-correlation.

2 Pipeline Overview

Harris responses are thresholded relative to the global maximum and then passed through a dilation-based NMS with a configurable radius, which spreads keypoints across the image instead of clustering them on the strongest corner. Each patch descriptor is normalised to zero mean and unit standard deviation, which makes Euclidean distance and NCC equivalent up to a monotonic transform. RANSAC samples three matches per iteration, rejects collinear triples before fitting, and after the main loop re-fits the affine on all inliers of the best model keeping the refit only if it does not lose inliers. The panorama canvas is sized from the projected corners of image 2, and the overlap is blended with a 50/50 average.

3 Main Analytical Results (sorbo)

I have used the sorbo pair as a reference case. Harris detection finds 641 and 670 corners, reduced to 613 and 633 after border-adjacent keypoints are dropped. With a 21×21 descriptor and Euclidean distance the top 300 matches are kept. Figure 2 shows the inliers (green) concentrated on the central overlap where the bike rack and building facades are visible in both views. RANSAC with 3000 iterations and a 3-pixel threshold accepts 185 inliers and rejects 115, giving a mean squared residual of 0.275 px^2 and a mean alignment error of 0.447 px . The recovered transform is essentially a pure horizontal translation of 533 pixels:

$$T = \begin{bmatrix} 1.000 & 0.000 & 533.2 \\ 0.000 & 1.000 & 0.001 \end{bmatrix}.$$

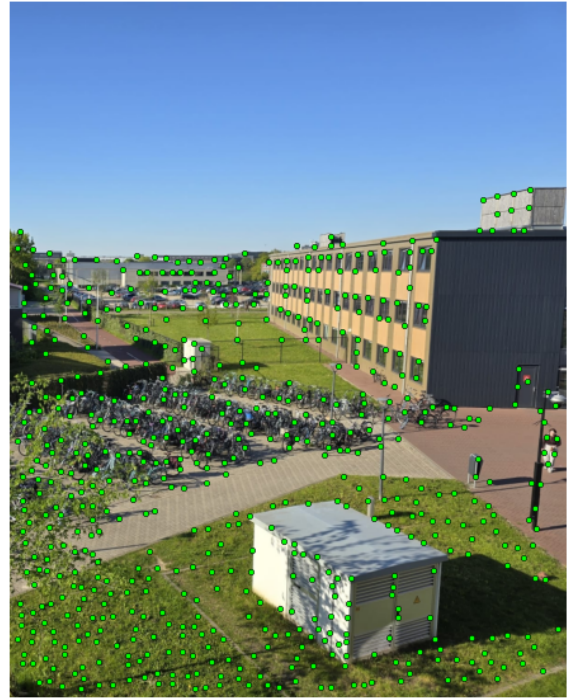
The panorama (Figure 3) reconstructs the full view; the faint vertical line in the middle is the blend seam.

Harris keypoints — image 1 (n=613)



(a) Image 1, 613 keypoints.

Harris keypoints — image 2 (n=633)



(b) Image 2, 633 keypoints.

Figure 1: Harris keypoints on the sorbo pair.

RANSAC: inliers (green) vs outliers (red) — inliers: 185 / 300

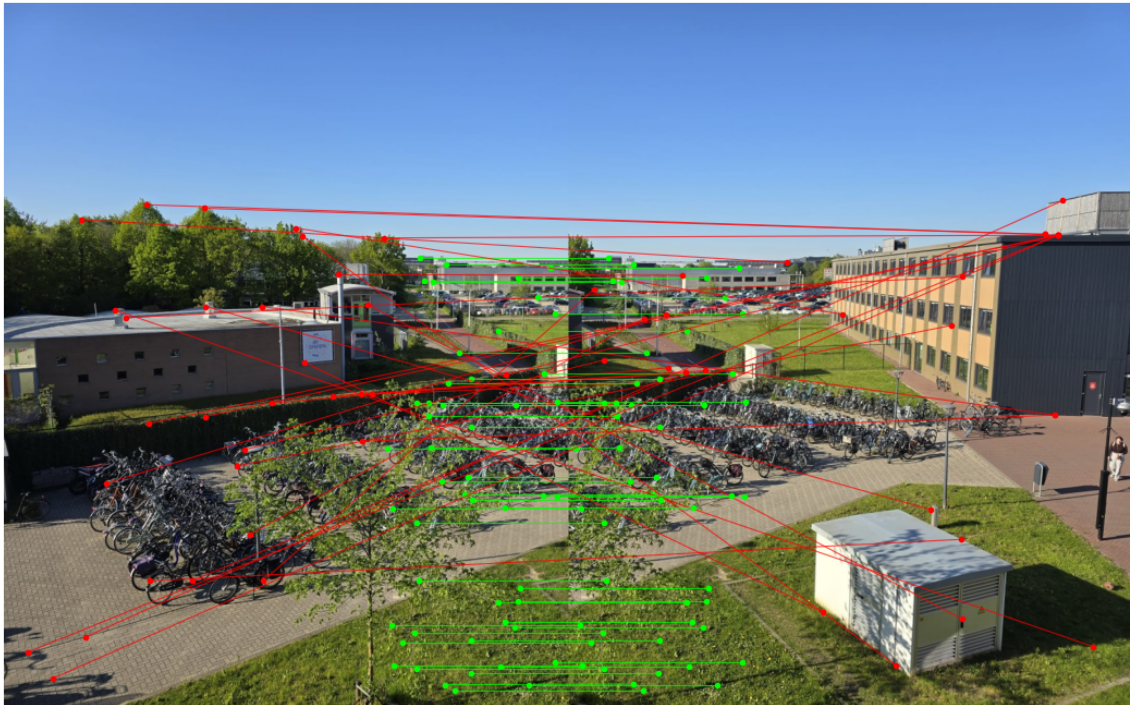


Figure 2: RANSAC inliers (green) and outliers (red)

Stitched panorama — sorbo



Figure 3: Final panorama for the sorbo pair.

4 Sensitivity Analyses

All sweeps are on the sorbo pair. I read the accuracy score as a mean alignment error in pixels, lower is better, but it has to be read jointly with the inlier count, because a setting that keeps only the very best matches will always report a low error.

4.1 Descriptor patch size

Sizes from 9×9 to 41×41 were tested (Figure 4). The inlier count peaks at size 15 (186) and stays near that level through size 33, then drops at size 41; the error grows monotonically from 0.289 px at size 9 to 0.485 px at size 41. Small patches give a tighter alignment with fewer matches, large patches the opposite. Size 21 was adopted as a compromise near the inlier peak.

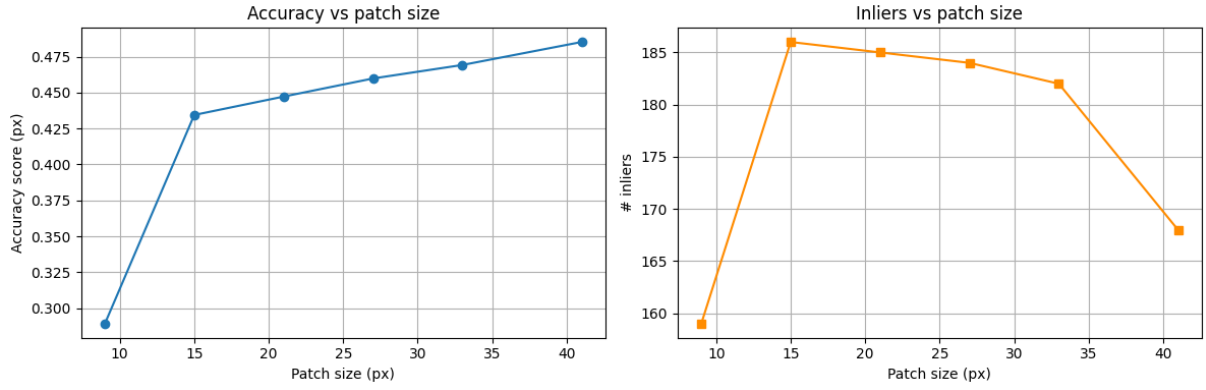


Figure 4: Alignment error and inlier count vs. patch size.

4.2 Number of retained matches (top- k)

Inliers grow from 50 at $k = 50$ to 185 at $k = 300$, and saturate at 192 from $k = 500$ onward (Figure 5). Beyond 300 the sorted distance list is almost exhausted and the extra candidates add very little. I kept $k = 300$.

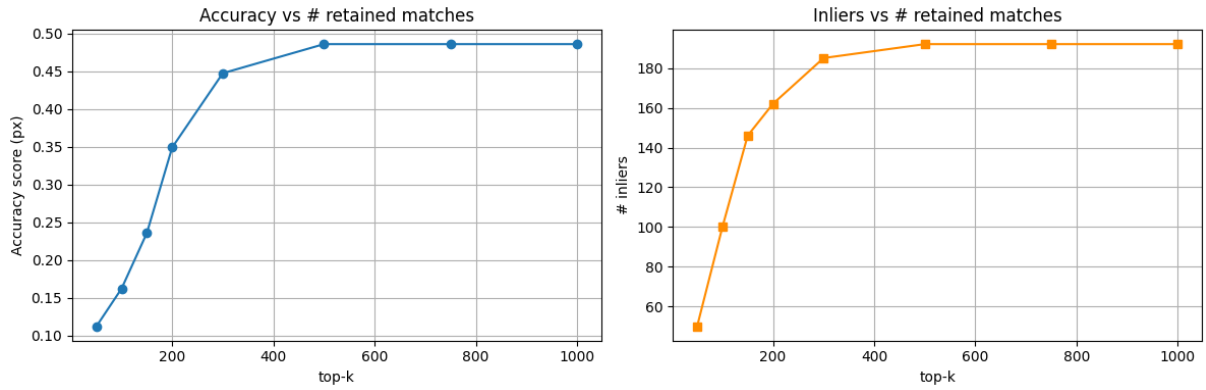


Figure 5: Effect of the top- k match selection parameter.

4.3 Descriptor matching threshold

Matches can also be filtered by an absolute distance threshold τ (6). The retained count grows almost linearly and the inlier count tracks it closely at low thresholds, before saturating around 190 matches, reaching 192 at $\tau \approx 1.2$: on this pair essentially every retained match is an inlier up to $\tau \approx 0.4$. Below $\tau \approx 0.2$ very few matches survive, sometimes approaching the minimum required for RANSAC. I preferred top- k because it does not depend on the absolute scale of descriptor distances.

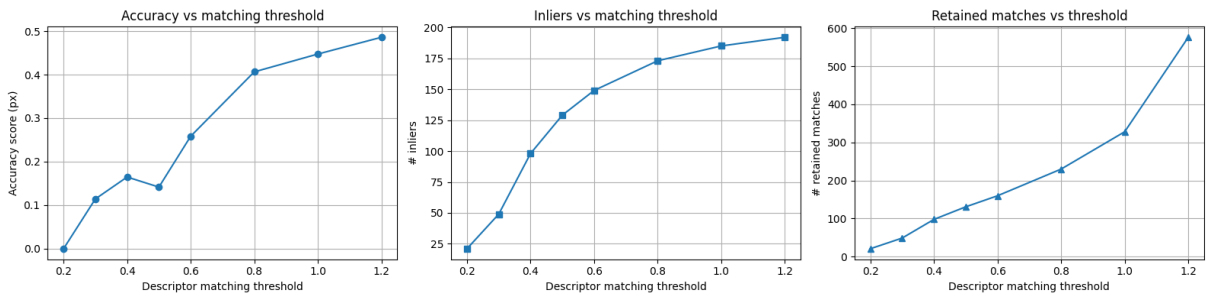


Figure 6: Effect of the descriptor matching threshold τ .

4.4 RANSAC iterations and inlier threshold

The inlier count is flat at 185 from 100 iterations onward (Figure 7): the scene is clean enough that even a small number of random triples finds the same consensus set. The inlier threshold shows a knee around 3 px: below, too many genuine matches are rejected; above, the numbers saturate. I chose 3000 iterations and a 3-px threshold.

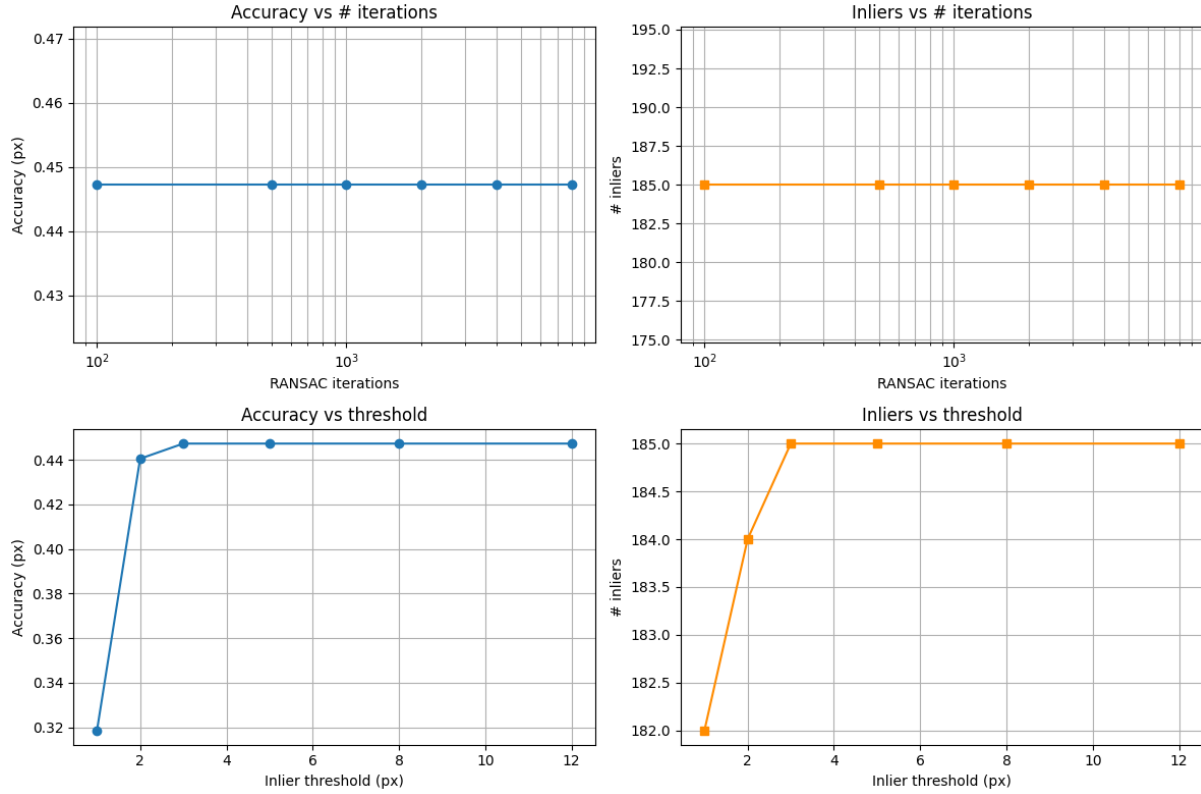


Figure 7: Effect of the number of RANSAC iterations (log scale) and effect of the RANSAC inlier threshold.

4.5 Euclidean vs. NCC

The two metrics gave identical results: 185 inliers and 0.447 px of error. Because my descriptors are pre-normalised to zero mean and unit standard deviation, Euclidean distance and NCC rank neighbours in the same order, so they pick the same matches.

5 Experiments on Additional Pairs

All three extra pairs were built by splitting a single wide photograph I took into two halves with $\sim 33\%$ overlap, which guarantees parallel geometry. Each pair targets a specific challenge.

Pair 1: Library bookshelf (repetitive texture). 821 / 641 keypoints after filtering, 155 inliers of 300, residual 2.25 px^2 , error 1.30 px (Figures 8, 9). The error is the largest among the three splits because many book spines look alike, so some inliers land one or two spines off.

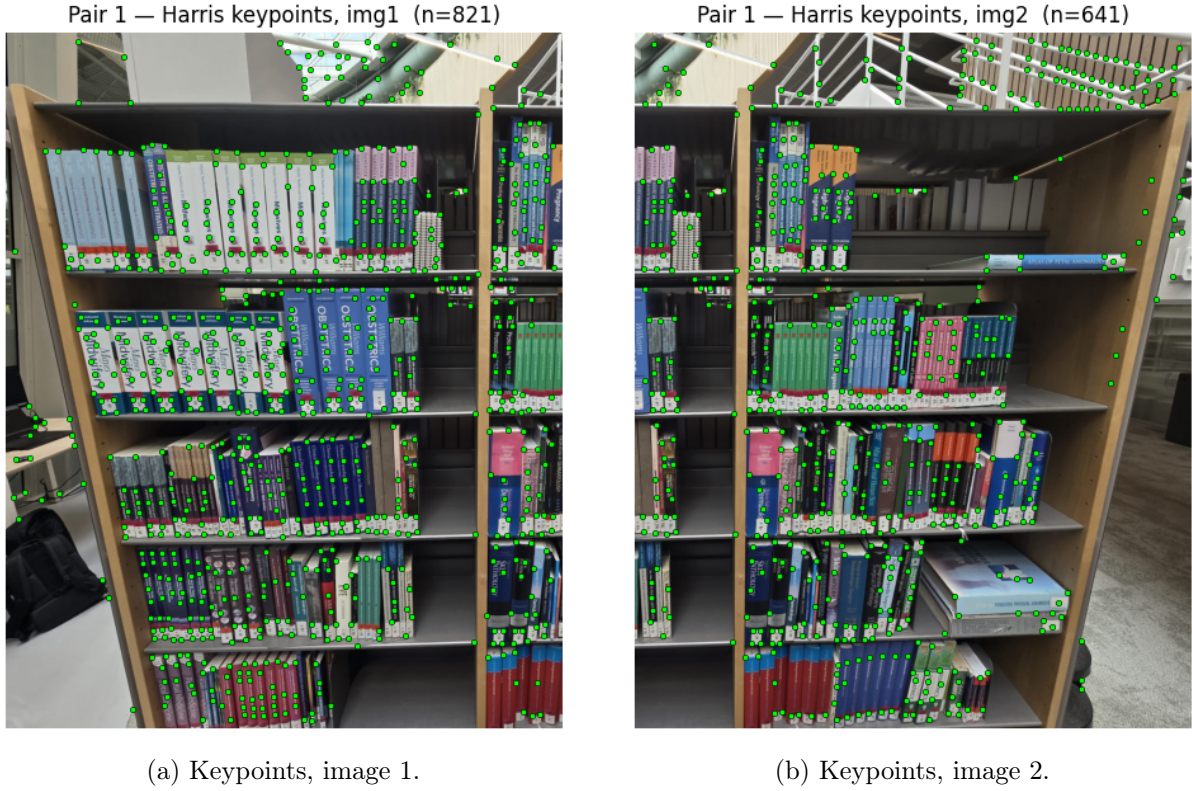


Figure 8: Pair 1 — Harris keypoints.



Figure 9: Pair 1 — RANSAC result and panorama.

Pair 2: Industrial night scene (low light, dark sky). 57 inliers of 170, residual 0.14 px², error 0.25 px (Figures 10, 11). Keypoints cluster in the lower half where the parking lot provides texture; the dark sky above produces almost none, which is exactly how Harris should behave on uniform regions.

Pair 2 — Harris keypoints, img1 (n=170)



(a) Keypoints, image 1.

Pair 2 — Harris keypoints, img2 (n=186)



(b) Keypoints, image 2.

Figure 10: Pair 2 — Harris keypoints.

Pair 2 — RANSAC inliers (green) / outliers (red) — inliers: 57 / 170



(a) Inliers / outliers.

Panorama — Pair 2

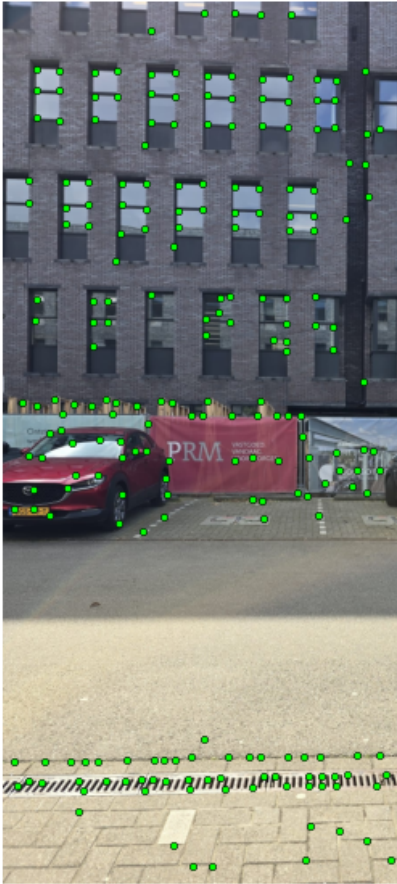


(b) Panorama.

Figure 11: Pair 2 — RANSAC result and panorama.

Pair 3: Office facade (periodic windows). 66 inliers of 228, residual 0.015 px^2 , error 0.035 px (Figures 12, 13). Inliers cluster on distinctive banners and cars; the windows themselves produce mostly outliers. When distinctive content is present, RANSAC recovers the right transform even when most of the image produces bad matches.

Pair 3 — Harris keypoints, img1 (n=228)



(a) Keypoints, image 1.

Pair 3 — Harris keypoints, img2 (n=228)



(b) Keypoints, image 2.

Figure 12: Pair 3 — Harris keypoints.

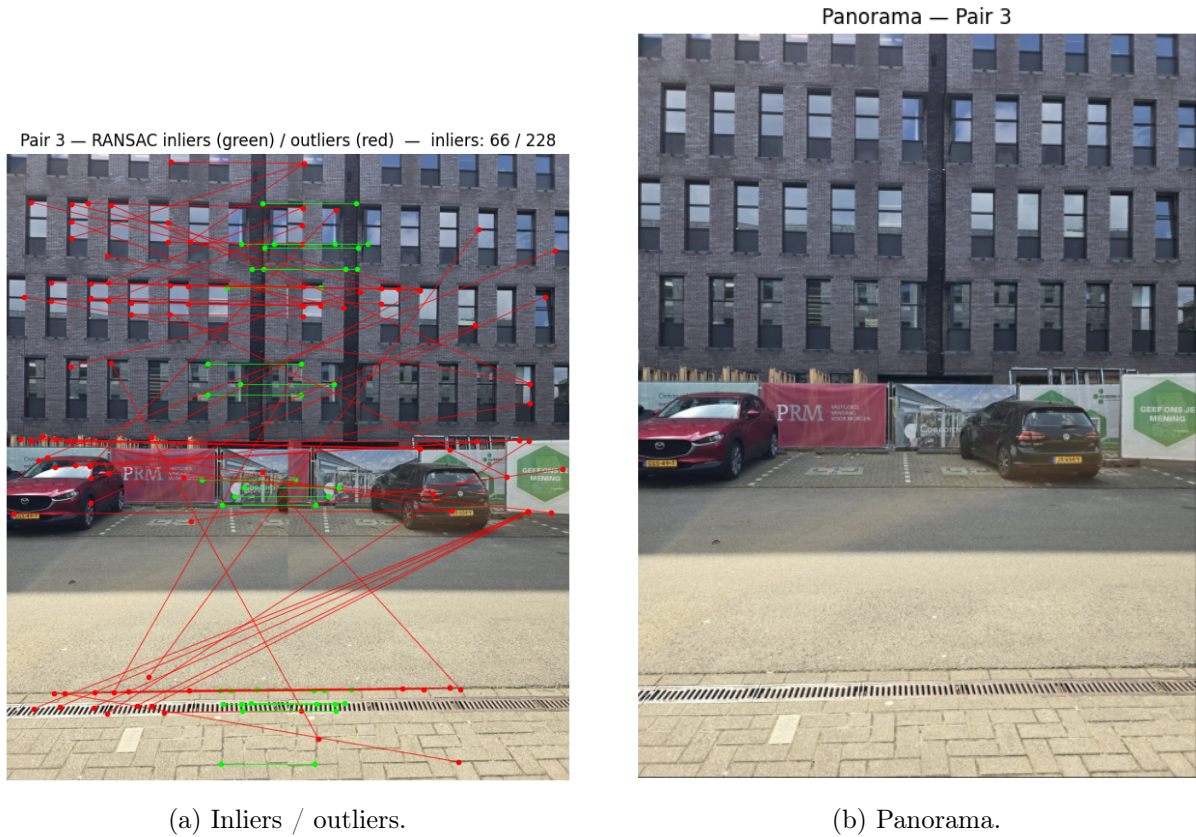


Figure 13: Pair 3 — RANSAC result and panorama.

Pair	#in	#out	residual (px ²)	error (px)
sorbo	185	115	0.275	0.447
Bookshelf	155	145	2.251	1.295
Industrial	57	113	0.144	0.246
Office	66	162	0.015	0.035

Table 1: Final results across all four pairs with the parameters selected in Section 4.

6 SIFT Descriptors on Harris Keypoints

As an optional extension I replaced the patch descriptor with SIFT (`cv2.SIFT_create`) at the same Harris keypoints. On the sorbo pair SIFT gave 165 inliers of 300, residual 1.009 px², error 0.845 px — fewer inliers and a worse alignment than the patch descriptor, but a clean panorama (Figure 14). On a scene with this much rigid texture the patch descriptor already captures the local structure well, so SIFT’s extra invariances cost accuracy rather than add it. I kept the patch descriptor as the default because it was already good enough on all four pairs.

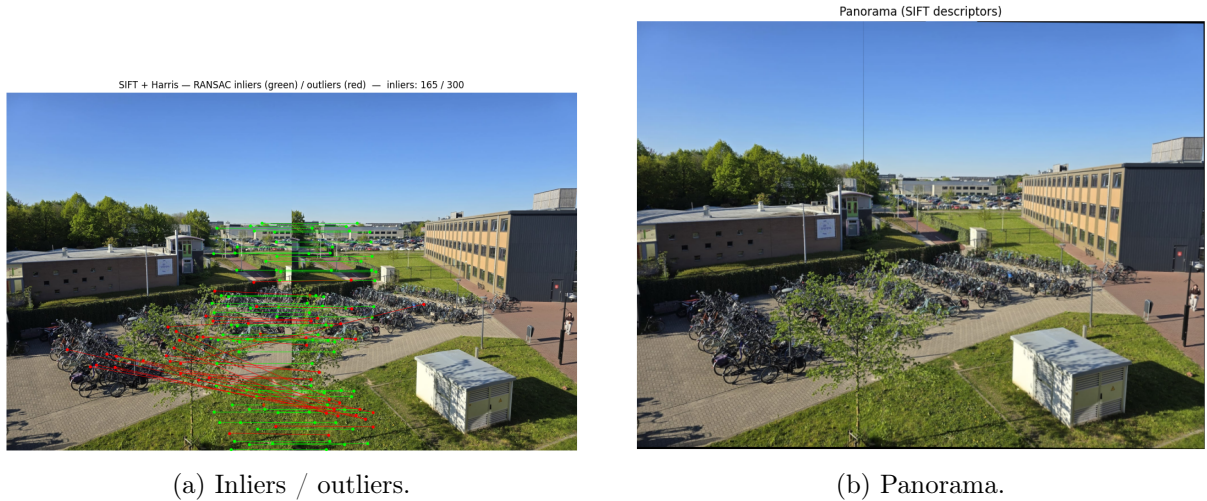


Figure 14: SIFT descriptors on Harris keypoints, sorbo pair.

7 Challenges and Observations

The hardest part of this assignment was dealing with silent failures. Early on, I tested the pipeline on photos that looked different to me but shared $\sim 85\%$ of their content. RANSAC returned a transform close to the identity, reported hundreds of “inliers” and a residual of exactly zero, and the panorama was just one image pasted on top of the other. What I learned is that the residual alone is not a quality indicator: near-zero means either a perfect fit or a degenerate one, and only the spread and geometry of the inliers separate the two.

Patch descriptors also struggle with periodic content. On the bookshelf and the office facade many descriptor matches connect visually similar but wrong points; RANSAC survives only because a small set of distinctive regions (banners, cars, labels) provides enough geometric signal to define the correct transform. Without those anchors the pipeline would have locked onto a shifted by one window transform and the numbers alone would not have revealed the failure.

8 Chosen Parameters and Conclusion

Based on the sweeps in Section 4 I settled on patch size 21×21 , $k = 300$, Euclidean distance, 3000 RANSAC iterations and a 3-pixel inlier threshold, producing the results in Table 1. The main failure modes were periodic texture and large uniform regions, both inherent limits of patch descriptors rather than bugs.